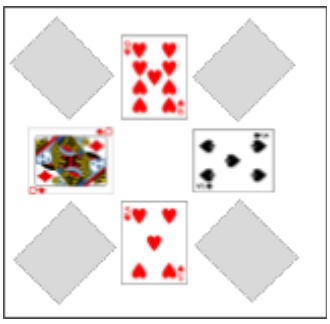
American Computer Science League

2025-2026 ● Contest 3: Kings in the Corners ● Intermediate Division



PROBLEM: *Kings in the Corners* is a two-person card game that uses a standard deck of 52 playing cards. The game begins by giving 7 cards to each player and placing four cards face-up creating 4 *original* piles. There are four *corner* piles that start empty. The remaining cards form the draw pile.

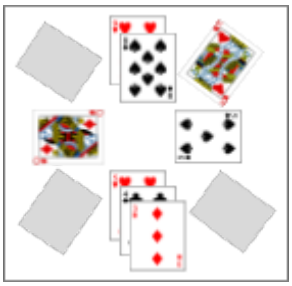
On a player's turn, a player attempts to play cards from his hand onto one of the 8 piles, each time, examining cards from left to right. The first playable card is moved according to these rules:

- **If the card is a King,** it is played onto the next available corner pile, starting with the upper right corner and moving clockwise.

- **If the card is not a King,** it can be played on an existing pile if it is the opposite color and exactly one rank lower than the top card of that pile. For example, the 4 of Clubs can be played on a 5 of Hearts or on a 5 of Diamonds.

Examine the existing piles in clockwise order, starting with the topmost.

When no more cards can be played, the turn ends, and the player draws a single card from the draw pile adding it as the last card in his hand. The game ends when no more cards can be played from either player's hand and the draw pile is empty. At the end of the game, output the top card of each of the 8 piles, starting at the topmost pile and proceeding clockwise. If a pile is empty, print "E". Separate each card with a single space.



Each card is represented by a 2-character string: *rank* followed by *suit*. The ranks, from high to low, are: K (king), Q (queen), J (jack), T (ten), 9, 8, 7, 6, 5, 4, 3, 2, and A (ace). The suits are C (clubs), D (diamonds), H (hearts), and S (spades). Clubs and Spades are black; Diamonds and Hearts are red.

The image at the left shows the board after playing 4 cards: 4C, KH, 8S, 3D. This does not match the example below.

EXAMPLE:

Input	Player 1: `JC 5C 7D QD KC 8S KS` Player 2: `7H 9S 3S 4H TH QH 6S` Draw Pile: `6C 7S 6H 9H AC 5D 2H QS KH TD JS`
Output	`5D 9S 7S TD AC QS 6S E`
Explanation	Cards are played as follows:

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Player:	Card Played:	Player's Hand:	Top Card on Each Pile:
Player 1	5C KC QD JC 8S 7D KS	JC 7D QD KC 8S KS JC 7D QD 8S KS JC 7D 8S KS 7D 8S KS 7D KS KS Draw AC	6C E 7S E **5C** E 9H E 6C **KC** 7S E 5C E 9H E 6C **QD** 7S E 5C E 9H E 6C **JC** 7S E 5C E 9H E 6C JC 7S E 5C E **8S** E 6C JC 7S E 5C E **7D** E 6C JC 7S **KS** 5C E 7D E
Player 2	4H 3S TH 9S QH 6S	7H 9S 3S TH QH 6S 7H 9S TH QH 6S 7H 9S QH 6S 7H QH 6S 7H 6S 7H, Draw 5D	6C JC 7S KS **4H** E 7D E 6C JC 7S KS **3S** E 7D E 6C **TH** 7S KS 3S E 7D E 6C **9S** 7S KS 3S E 7D E 6C 9S 7S **QH** 3S E 7D E 6C 9S 7S QH 3S E **6S** E
Player 1	none	AC, Draw 2H	6C 9S 7S QH 3S E 6S E
Player 2	5D	7H, Draw QS	**5D** 9S 7S QH 3S E 6S E
Player 1	2H AC	AC Draw KH	5D 9S 7S QH **2H** E 6S E 5D 9S 7S QH **AC** E 6S E
Player 2	none	7H QS, Draw TD	5D 9S 7S QH AC E 6S E
Player 1	KH	Draw JS	5D 9S 7S QH AC **KH** 6S E
Player 2	QS	7H TD, Empty ple	5D 9S 7S QH AC **QS** 6S E
Player 1	JS	Empty pile	5D 9S 7S **JS** AC QS 6S E
Player 2	TD	7H, Empty pile	5D 9S 7S **TD** AC QS 6S E
		No more cards to play	

Since neither player can play a card and the draw pile is empty, the game ends. Output the top card of each of the eight piles. The output is: 5D 9S 7S TD AC QS 6S E

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INPUT: Input two strings representing two players' hands and a string representing the draw pile. The first 4 cards of the draw pile are placed face-up to create 4 original piles to start the game. Cards are represented as 2-character strings, each separated by a single space.

OUTPUT: After the game is over, output a string representing the top card on each of the 8 piles using "E" if it is empty, each separated by a single space.

SAMPLE INPUT:

1. JC 5C 7D QD KC 8S KS
   7H 9S 3S 4H TH QH 6S
   6C 7S 6H 9H AC 5D 2H QS KH TD JS
2. AD 5C 2S TH 2H 3C QS
   6S 9D 8H KH TD KS 5D
   JH 7D QH 3D 7H 2C 6H 8C QD JC 4S AC AS 9C 7S JS 3S 8D AH 8S 4D 6C
3. 3D JH 2C 2S KS AH 8S
   5H QS 5S 5D KD 3H 6D
   9H 4C JD 2D QH 3S JC 3C 4S 9S JS AC QC 9C 6C 4D TH
4. 7S JD 9S 8C 8S 9D 4C
   KH TH KS 7H KC 3S TC
   AH 7D 8D 8H 2D 6S 6H QC 5D 5S 9C JS QH QS 2H AD
5. 9D 9C AH 8H KH 3C 2S
   2D TS JC 5S 6C TD 8C
   AS AD 6S QH 3H 8D QS 4S 8S 5H 6D 3D 7D JH 2C 2H
6. 5S 8D 5D AC 8S 2H QH
   JH 2S 9S KS KH 8H 7S
   4S 2D JD TH TD AD 5C 7D 6C 9D 6S 3C 3S 6H QS 9H 4C TC 7H 4H 2C JC TS

SAMPLE OUTPUT:

1. 5D 9S 7S TD AC QS 6S E
2. JH QS 4S AS TH E AD E
3. 8S 9S AH JH JD E AC E
4. AH 7H 4C 9S 5S KC 8H E
5. AS 6C AD E AH E 8H E
6. 4S TD AC 4C 9H E 2H E

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TEST DATA

TEST INPUT:

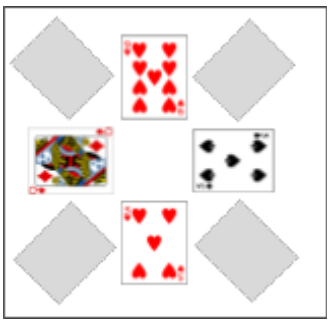
1. TC 5S 8H JH KD 4D 9C
   2H 3D JC AH QD 8S 8C
   TD 6S 9D 6C AS 4H 8D 3C 2S 5H QC JD 7C 7D 4C 6D TH 2D 2C
2. 6C KH 7H 9D 6D 7S 9C
   4D 8H TS 5H JD QS 3H
   TD 4S 8D 7C JH 9S AH TC 5S 4H 8C 6S JC 2S 3C 5C
3. 2C QD JH AD QC 8H KC
   9H QS 9D KD 5C 6S JC
   7C 7D 5D 4H 8S 2H QH 3S AC 3H 4D 6D 5H TC TS JD JS AH 6C
4. KH 4H 9H 2D 2H AS 3S
   KS TD TS 8C 7S QD 9D
   3C AC JS QS AH 8D JH QH 7D 6H 8S 6S 7C 9C TC 7H 5C
5. KD TD AS 6C 8S 2H 4D
   3H 5H 4S 8D 6S 5C 7H
   JS 7S 5D 2S JD 6H JH QS AH 9C 9H 3S 7C 3D AD TS TC 4H 7D 3C 8H
6. 6C 7H 9S 2C KC 5C 5D
   8H JS AH KD 7S TD QS
   JC 4H 9H QH 8C 6H 7C 3S 4D JH 7D 4C 9D 3D AC TC 8D 8S AD 3C

TEST OUTPUT:

1. AS TC AH E 7D E 6C E
2. 8H 5H AH E 7S E 3C E
3. 4D JC 5H 8S 5D E AC E
4. AS KH AC QD 2H E 6S E
5. 7C 5H AS E 3H E AH E
6. 3C KC 3S 7D AH E JS E

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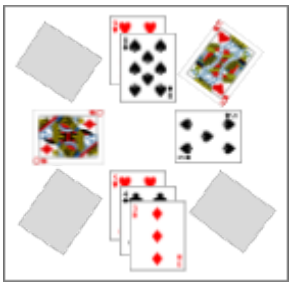
问题: *角落国王* 是一款双人纸牌游戏，使用一副标准的 52 张扑克牌。游戏开始时，每名玩家各发 7 张牌，并面朝上放置四张牌，形成四个*初始牌堆*。另有四个*角落牌堆* 初始为空。剩余的牌组成抽牌堆。

轮到玩家回合时，该玩家需将自己手中的牌放到 8 个牌堆中的其中一个。每次操作时，玩家从左到右检查自己手中的牌。第一张可以打出的牌遵循一下规则：

- **如果这张牌是 K（国王）**
  将其放入下一个可用的角落牌堆。角落牌堆按顺时针顺序填充，从右上角开始。
- **如果这张牌不是 K**
  这张牌可以放入一个现有牌堆，条件是这张牌的颜色与牌堆顶牌的颜色相反，且牌面点数比顶牌正好小 1。例如，梅花 4 可以放在红桃 5 或方块 5 的上面。

按顺时针顺序检查各个现有牌堆，从最上方的牌堆开始。

当无法再打出任何牌时，该回合结束，玩家从抽牌堆中抽一张牌，将其作为手中的最后一张牌。游戏在所有玩家都无法打出任何牌且抽牌堆为空时结束。游戏结束时，请按照顺时针顺序，从最上方的牌堆开始，依次输出 8 个牌堆的顶牌。如果某个牌堆为空，则输出“ E ”。每张牌之间用一个空格分隔。



每张牌用两个字符的字符串表示：*点数*后跟*花色*。点数从高到低依次为：K（国王）、Q（皇后）、J（杰克）、T （10）、9、8、7、6、5、4、3、2、A 牌。花色分别为：C（梅花）、D（方块）、H（红桃）、S（黑桃）。梅花和黑桃为黑色；方块和红桃为红色。

左图展示了按照 梅花 4、红桃 K、黑桃 8、方块 3 的顺序打出四张牌后的牌局情况。

示例：

输入	玩家 1: `JC 5C 7D QD KC 8S KS` 玩家 2: `7H 9S 3S 4H TH QH 6S` 抽牌**堆**: `6C 7S 6H 9H AC 5D 2H QS KH TD JS`
输出	`5D 9S 7S TD AC QS 6S E`
解释说明	**出牌规则如下：**

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玩家:	已出的牌:	玩家手中的牌:	各牌堆顶牌：
玩家1	5C KC QD JC 8S 7D KS	JC 7D QD KC 8S KS JC 7D QD 8S KS JC 7D 8S KS 7D 8S KS 7D KS KS Draw AC	6C E 7S E **5C** E 9H E 6C **KC** 7S E 5C E 9H E 6C **QD** 7S E 5C E 9H E 6C **JC** 7S E 5C E 9H E 6C JC 7S E 5C E **8S** E 6C JC 7S E 5C E **7D** E 6C JC 7S **KS** 5C E 7D E
玩家2	4H 3S TH 9S QH 6S	7H 9S 3S TH QH 6S 7H 9S TH QH 6S 7H 9S QH 6S 7H QH 6S 7H 6S 7H, Draw 5D	6C JC 7S KS **4H** E 7D E 6C JC 7S KS **3S** E 7D E 6C **TH** 7S KS 3S E 7D E 6C **9S** 7S KS 3S E 7D E 6C 9S 7S **QH** 3S E 7D E 6C 9S 7S QH 3S E **6S** E
玩家1	none	AC, Draw 2H	6C 9S 7S QH 3S E 6S E
玩家2	5D	7H, Draw QS	**5D** 9S 7S QH 3S E 6S E
玩家1	2H AC	AC Draw KH	5D 9S 7S QH **2H** E 6S E 5D 9S 7S QH **AC** E 6S E
玩家2	none	7H QS, Draw TD	5D 9S 7S QH AC E 6S E
玩家1	KH	Draw JS	5D 9S 7S QH AC **KH** 6S E
玩家2	QS	7H TD, Empty ple	5D 9S 7S QH AC **QS** 6S E
玩家1	JS	Empty pile	5D 9S 7S **JS** AC QS 6S E
玩家2	TD	7H, Empty pile	5D 9S 7S **TD** AC QS 6S E
		No more cards to play	

由于两名玩家都无牌可出，且抽牌堆已空，游戏结束。

此时输出八个牌堆各自的顶牌。输出结果为: 5D 9S 7S TD AC QS 6S E

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输入: 输入共三行字符串，前两行分别代表两位玩家的手牌，第三行代表抽牌堆。游戏开始时，抽牌堆的前四张牌将面朝上放置，形成四个初始牌堆。每张牌用两个字符的字符串表示，每张牌之间用一个空格分隔。

输出: 游戏结束后，输出一行字符串，表示 8 个牌堆各自的顶牌，如果某个牌堆为空则用“E”表示，每张牌之间用一个空格分隔。

样例输入:

1. JC 5C 7D QD KC 8S KS
   7H 9S 3S 4H TH QH 6S
   6C 7S 6H 9H AC 5D 2H QS KH TD JS
2. AD 5C 2S TH 2H 3C QS
   6S 9D 8H KH TD KS 5D
   JH 7D QH 3D 7H 2C 6H 8C QD JC 4S AC AS 9C 7S JS 3S 8D AH 8S 4D 6C
3. 3D JH 2C 2S KS AH 8S
   5H QS 5S 5D KD 3H 6D
   9H 4C JD 2D QH 3S JC 3C 4S 9S JS AC QC 9C 6C 4D TH
4. 7S JD 9S 8C 8S 9D 4C
   KH TH KS 7H KC 3S TC
   AH 7D 8D 8H 2D 6S 6H QC 5D 5S 9C JS QH QS 2H AD
5. 9D 9C AH 8H KH 3C 2S
   2D TS JC 5S 6C TD 8C
   AS AD 6S QH 3H 8D QS 4S 8S 5H 6D 3D 7D JH 2C 2H
6. 5S 8D 5D AC 8S 2H QH
   JH 2S 9S KS KH 8H 7S
   4S 2D JD TH TD AD 5C 7D 6C 9D 6S 3C 3S 6H QS 9H 4C TC 7H 4H 2C JC TS

样例输出:

1. 5D 9S 7S TD AC QS 6S E
2. JH QS 4S AS TH E AD E
3. 8S 9S AH JH JD E AC E
4. AH 7H 4C 9S 5S KC 8H E
5. AS 6C AD E AH E 8H E
6. 4S TD AC 4C 9H E 2H E

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测试数据

测试输入:

```
1. TC 5S 8H JH KD 4D 9C
   2H 3D JC AH QD 8S 8C
   TD 6S 9D 6C AS 4H 8D 3C 2S 5H QC JD 7C 7D 4C 6D TH 2D 2C
2. 6C KH 7H 9D 6D 7S 9C
   4D 8H TS 5H JD QS 3H
   TD 4S 8D 7C JH 9S AH TC 5S 4H 8C 6S JC 2S 3C 5C
3. 2C QD JH AD QC 8H KC
   9H QS 9D KD 5C 6S JC
   7C 7D 5D 4H 8S 2H QH 3S AC 3H 4D 6D 5H TC TS JD JS AH 6C
4. KH 4H 9H 2D 2H AS 3S
   KS TD TS 8C 7S QD 9D
   3C AC JS QS AH 8D JH QH 7D 6H 8S 6S 7C 9C TC 7H 5C
5. KD TD AS 6C 8S 2H 4D
   3H 5H 4S 8D 6S 5C 7H
   JS 7S 5D 2S JD 6H JH QS AH 9C 9H 3S 7C 3D AD TS TC 4H 7D 3C 8H
6. 6C 7H 9S 2C KC 5C 5D
   8H JS AH KD 7S TD QS
   JC 4H 9H QH 8C 6H 7C 3S 4D JH 7D 4C 9D 3D AC TC 8D 8S AD 3C
```

测试输出:

```
1. AS TC AH E 7D E 6C E
2. 8H 5H AH E 7S E 3C E
3. 4D JC 5H 8S 5D E AC E
4. AS KH AC QD 2H E 6S E
5. 7C 5H AS E 3H E AH E
6. 3C KC 3S 7D AH E JS E
```